BioFraMe - Hydrogenation catalysts based on bio-based deep eutectic solvents

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Catalysts play a key role in the conversion of renewable raw materials into basic chemicals for the chemical industry, e.g. through hydrogenation. Trace amounts of impurities in the educt can lead to a loss of activity during hydrogenation due to ageing and coking, meaning that the catalysts have to be replaced periodically.

To solve these problems, novel recyclable catalysts have been developed using deep eutectic solvents (DES) based on renewable raw materials as carriers and metal oxides as the starting material for the active component. Urea and sugar derivatives, for example, can be used as DES.

The catalyst is produced by a novel synthesis at temperatures of approx. 400 °C. The raw material is then activated under reductive conditions, in which the metal oxide is reduced to catalytically active metal. In order to optimize this sub-process, tests were carried out in a magnetic suspension balance from Linseis Messgeräte GmbH. The activation is balanced by elemental analysis before and after the reaction and the change in mass during the reduction, which can be precisely recorded using the balance.

In order to test the catalytic activity of the nickel-based catalysts, studies were carried out on the hydrogenation of cinnamic acid to phenylpropanoic acid and a comparison was made with conventional carbon-supported material.

The catalysts are less active than conventional catalysts, but have advantages due to the simple and cost-effective production technology and the recyclability of the metal component.

The production technology, activation and results for testing the catalytic activity of nickel catalysts produced using mixtures of vanillin, choline chloride, dimetylurea, urea or sorbitol are described in detail.

During activation, a significant decrease in mass and a change in chemical composition can be observed in the glucose/urea-based materials, which can be explained by the following reactions:

- The oxygen content decreases during activation due to the conversion of nickel oxide to metallic nickel, whereby the reaction product water evaporates.
- However, the drop in the oxygen content is more pronounced than would be expected on the basis of the nickel oxide reduction, which indicates reactions in the matrix. It is likely that the glucose or its derivatives from the synthesis are further dehydrated. This water splitting reduces the oxygen content and the total mass.
- The decreasing nitrogen content indicates a decomposition of the urea structures in the matrix.

The activated materials were tested in hydrogenation processes. In the gas phase, carbon dioxide was converted with hydrogen at 450 °C to form methane. In the liquid phase, cinnamic acid was hydrogenated to phenylpropionic acid.

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